Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current voltage a part of a circuit (box N)

Review: equations from the graph alone

	incidence / mesh	**fundamental cutset / loop**
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	node voltages $\mathbf{e}$	tree voltages $\mathbf{v}_T$
$b-n+1$ currents	mesh currents $\mathbf{i}_M$	co-tree currents $\mathbf{i}_C$
works for	M: planar, nonseparable	any connected graph

KCL and KVL depend only on the graph: they hold for any elements.

A circuit has $2b$ unknowns: b more equations come from the elements.

Why this lecture?

- **So far:** KCL and KVL, written for any elements.

Why this lecture?

- **So far:** KCL and KVL, written for any elements.
- **Now:** LTI resistive circuits, where every equation is linear and algebraic.

Why this lecture?

- **So far:** KCL and KVL, written for any elements.
- **Now:** LTI resistive circuits, where every equation is linear and algebraic.
- **Two ways to solve:** transform the circuit into a simpler one, or write its equations as it is.

Why this lecture?

- **So far:** KCL and KVL, written for any elements.
- **Now:** LTI resistive circuits, where every equation is linear and algebraic.
- **Two ways to solve:** transform the circuit into a simpler one, or write its equations as it is.
- **This lecture — transform:** replace a part by a source, or by a simpler part with the same terminal behaviour
  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation

Why this lecture?

- **So far:** KCL and KVL, written for any elements.
- **Now:** LTI resistive circuits, where every equation is linear and algebraic.
- **Two ways to solve:** transform the circuit into a simpler one, or write its equations as it is.
- **This lecture — transform:** replace a part by a source, or by a simpler part with the same terminal behaviour
  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation
- **Later — the circuit as it is:** node voltage and mesh current analysis

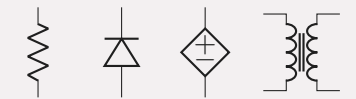
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

Resistive and dynamic elements

Resistive element: no memory

Its law relates v and i at the same instant, algebraically.

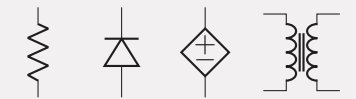


later: the ideal op-amp

Resistive and dynamic elements

Resistive element: no memory

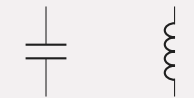
Its law relates v and i at the same instant, algebraically.



later: the ideal op-amp

Dynamic element: memory

Its law has a derivative.

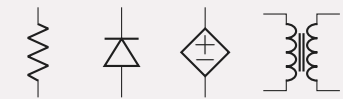


also coupled inductors

Resistive and dynamic elements

Resistive element: no memory

Its law relates v and i at the same instant, algebraically.



later: the ideal op-amp

Dynamic element: memory

Its law has a derivative.



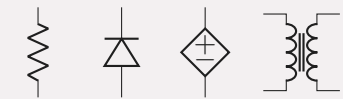
also coupled inductors

"Resistive" means memoryless, not only "a resistor".

Resistive and dynamic elements

Resistive element: no memory

Its law relates v and i at the same instant, algebraically.



later: the ideal op-amp

Dynamic element: memory

Its law has a derivative.



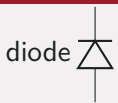
also coupled inductors

"Resistive" means memoryless, not only "a resistor".

An element is linear if its law is linear in v, i; time-invariant if its law does not change with time.

Three questions about a circuit

Linear?

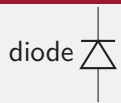


Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

Three questions about a circuit

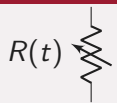
Linear?



Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

Time-invariant?

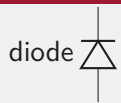


Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

Three questions about a circuit

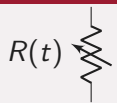
Linear?



Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

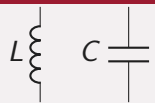
Time-invariant?



Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

Resistive?

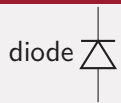


Dynamic if it contains at least one dynamic element: an L or a C.

Otherwise **resistive**.

Three questions about a circuit

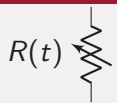
Linear?



Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

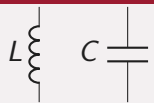
Time-invariant?



Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

Resistive?



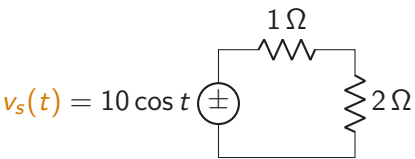
Dynamic if it contains at least one dynamic element: an L or a C.

Otherwise **resistive**.

LTI resistive circuit = linear + time-invariant + resistive:

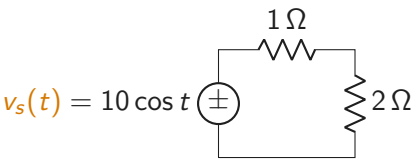
apart from the independent sources, every element is linear, time-invariant and resistive.

Independent sources are inputs



The source varies in time. Is the circuit time-varying?

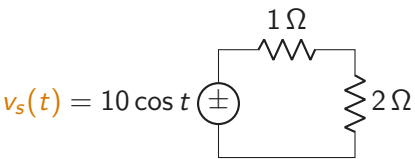
Independent sources are inputs



The source varies in time. Is the circuit time-varying?

No. An independent source is an external input: a given signal of time.

Independent sources are inputs



The source varies in time. Is the circuit time-varying?

No. An independent source is an external input: a given signal of time.

The three questions are about the circuit that the inputs drive, not about the inputs. This is an LTI resistive circuit.

What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

Time-invariant resistor

$$v(t) = R\, i(t)$$

the same rule at every instant

What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

Time-invariant resistor

$$v(t) = R\, i(t)$$

the same rule at every instant

Time-varying resistor

$$v(t) = R(t)\, i(t)$$

the rule itself changes with t

What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

Time-invariant resistor

$$v(t) = R\, i(t)$$

the same rule at every instant

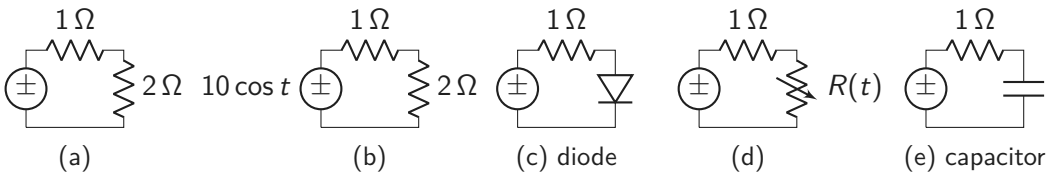
Time-varying resistor

$$v(t) = R(t)\, i(t)$$

the rule itself changes with t

Linear or nonlinear is also a property of the element law: $v = Ri$ is linear, a diode law is not.

Which circuit is which?

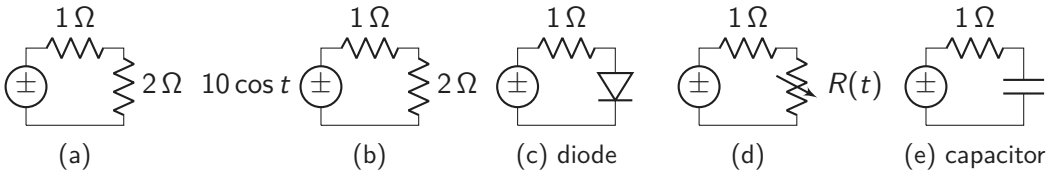


(a) (b) (c) diode (d) (e) capacitor

A source, a $1\,\Omega$ resistor and one more element. Sources: $5\,\mathrm{V}$ except (b). In (d), $R(t) = 2 + \sin t\ \Omega$.

Linear? Time-invariant? Resistive?

Which circuit is which?



(a) (b) (c) diode (d) (e) capacitor

	(a)	(b)	(c)	(d)	(e)
linear	✓	✓	×	✓	✓
time-invariant	✓	✓	✓	×	✓
resistive	✓	✓	✓	✓	×
LTI resistive	✓	✓	×	×	×

What we gain

Always — no assumption on the elements

KCL and KVL are linear for every circuit: they come from the graph alone.

What we gain

Always — no assumption on the elements

KCL and KVL are linear for every circuit: they come from the graph alone.

Because the circuit is LTI resistive

Every element equation is linear and algebraic, with no derivative:

$$v = Ri \quad \text{(resistor)}, \qquad v = \mu v_x, \quad i = g v_x, \quad \ldots \quad \text{(dependent sources)}$$

What we gain

Always — no assumption on the elements

KCL and KVL are linear for every circuit: they come from the graph alone.

Because the circuit is LTI resistive

Every element equation is linear and algebraic, with no derivative:

$$v = Ri \quad \text{(resistor)}, \qquad v = \mu v_x, \quad i = g v_x, \quad \ldots \quad \text{(dependent sources)}$$

- Time enters only through the values of the independent sources.

What we gain

Always — no assumption on the elements

KCL and KVL are linear for every circuit: they come from the graph alone.

Because the circuit is LTI resistive

Every element equation is linear and algebraic, with no derivative:

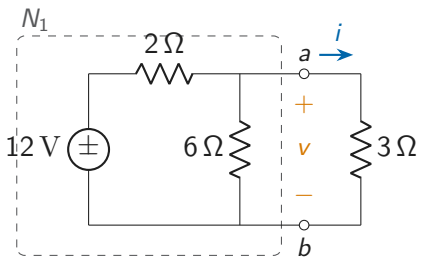
$$v = Ri \ \text{(resistor)}, \qquad v = \mu v_x, \quad i = g v_x, \ \ldots \ \text{(dependent sources)}$$

- Time enters only through the values of the independent sources.
- So at each instant the circuit is solved like a circuit with constant sources: $2b$ linear algebraic equations.

Roadmap

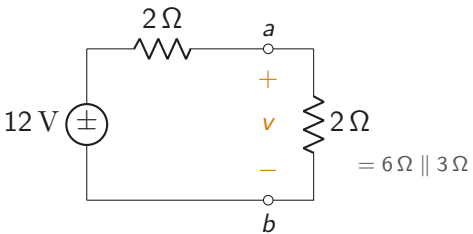
1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

A first example: simplify the circuit



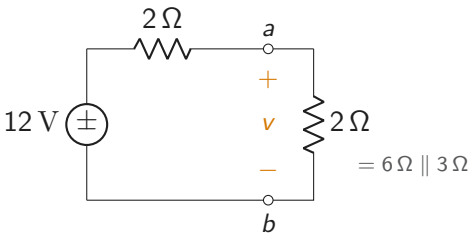
Find v and i.

A first example: simplify the circuit



Step 1. $6\,\Omega$ and $3\,\Omega$ are in parallel: together they act as one $2\,\Omega$.

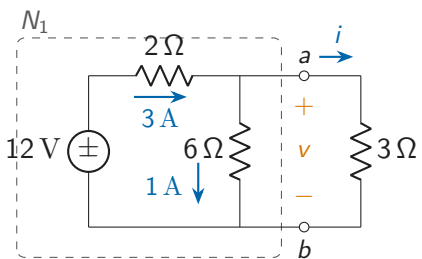
A first example: simplify the circuit



Step 1. $6\,\Omega$ and $3\,\Omega$ are in parallel: together they act as one $2\,\Omega$.

Step 2. The two $2\,\Omega$ share the $12\,\mathrm{V}$ equally: $v = 6\,\mathrm{V}$.

A first example: simplify the circuit



Step 1. $6\,\Omega$ and $3\,\Omega$ are in parallel: together they act as one $2\,\Omega$.

Step 2. The two $2\,\Omega$ share the $12\,\mathrm{V}$ equally: $v = 6\,\mathrm{V}$.

Step 3. Back to the original circuit: $i = 6/3 = 2\,\mathrm{A}$; the $2\,\Omega$ carries $3\,\mathrm{A}$, the $6\,\Omega$ carries $1\,\mathrm{A}$.

What did we just do?

- We replaced a part ($6\,\Omega \parallel 3\,\Omega$) by a simpler part ($2\,\Omega$).

What did we just do?

- We replaced a part ($6\,\Omega \parallel 3\,\Omega$) by a simpler part ($2\,\Omega$).
- The rest of the circuit did not notice: v stayed the same.

What did we just do?

- We replaced a part ($6\,\Omega \parallel 3\,\Omega$) by a simpler part ($2\,\Omega$).
- The rest of the circuit did not notice: v stayed the same.
- Knowing v, we went back and found the values inside.

What did we just do?

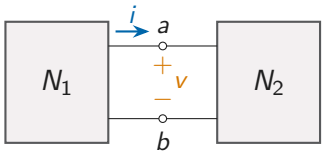
- We replaced a part ($6\,\Omega \parallel 3\,\Omega$) by a simpler part ($2\,\Omega$).
- The rest of the circuit did not notice: v stayed the same.
- Knowing v, we went back and found the values inside.

question	answered by
May a part be replaced by a source?	substitution theorem
When may a part be replaced by a simpler part?	equivalent one-ports
Which simple replacements are there?	series, parallel, source transformation

Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

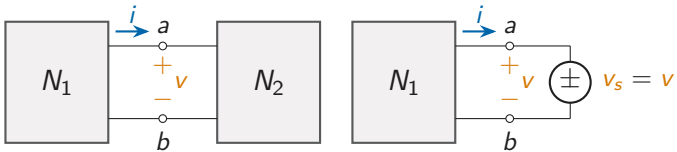
The substitution theorem



(I)

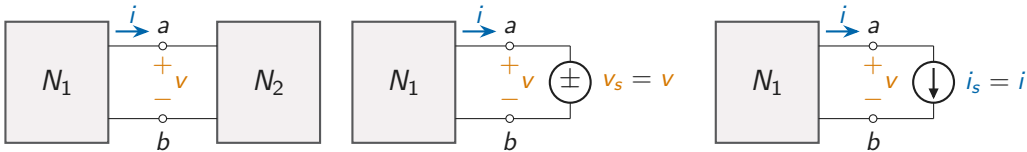
A circuit cut into two parts, joined at the terminals a, b.
i: current from N_1 into N_2 v: voltage between a and b

The substitution theorem



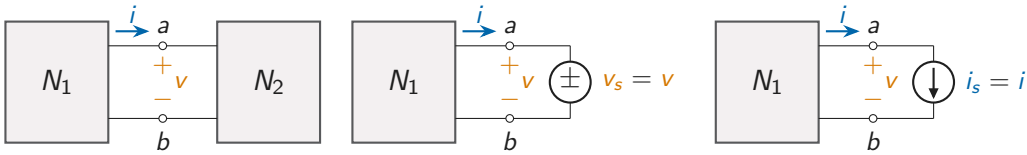
Replace N_2 by a voltage source with $v_s(t) = v(t)$ ...

The substitution theorem



... or by a current source with $i_s(t) = i(t)$.

The substitution theorem



Substitution theorem

No branch voltage or branch current inside N_1 changes. An observer inside N_1 cannot tell (I), (II) and (III) apart.

Why does it work?

- The equations inside N_1 (KCL, KVL, element equations) do not change.

Why does it work?

- The equations inside N_1 (KCL, KVL, element equations) do not change.
- N_1 meets the rest only through v and i.

Why does it work?

- The equations inside N_1 (KCL, KVL, element equations) do not change.
- N_1 meets the rest only through v and i.
- The source fixes one of them at the value it had in (I).

Why does it work?

- The equations inside N_1 (KCL, KVL, element equations) do not change.
- N_1 meets the rest only through v and i.
- The source fixes one of them at the value it had in (I).
- So the solution of (I) still satisfies every equation of (II) and (III).

Why does it work?

- The equations inside N_1 (KCL, KVL, element equations) do not change.
- N_1 meets the rest only through v and i.
- The source fixes one of them at the value it had in (I).
- So the solution of (I) still satisfies every equation of (II) and (III).
- N_2 can be anything, even nonlinear: only its port values matter.

Why does it work?

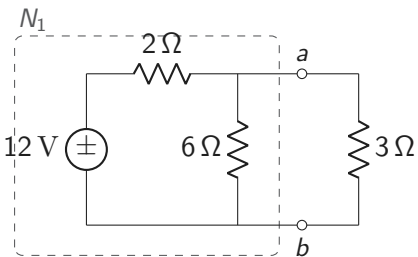
- The equations inside N_1 (KCL, KVL, element equations) do not change.
- N_1 meets the rest only through v and i.
- The source fixes one of them at the value it had in (I).
- So the solution of (I) still satisfies every equation of (II) and (III).
- N_2 can be anything, even nonlinear: only its port values matter.

One condition

The new circuit must have a unique solution.

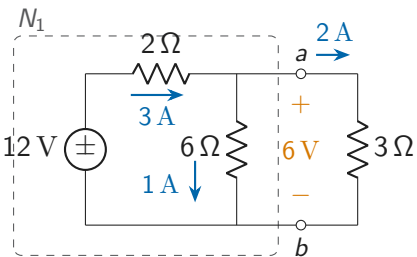
Example: N_1 a single current source and N_2 replaced by a current source $\Rightarrow$ v undetermined.

Example: substitution



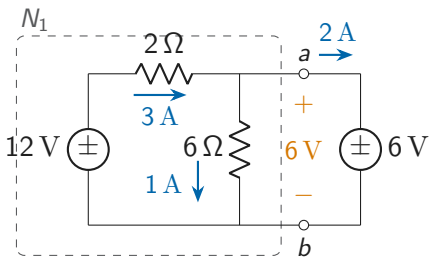
The first example, cut at a, b: N_2 is the $3\,\Omega$ resistor.

Example: substitution



(I) From the first example: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; inside N_1: $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

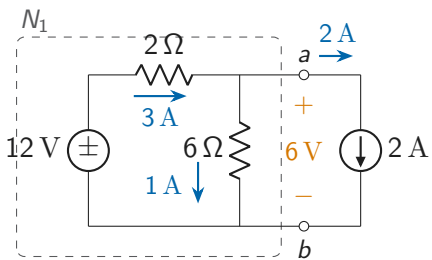
Example: substitution



(I) From the first example: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; inside N_1: $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

(II) A $6\,\mathrm{V}$ source fixes $v = 6\,\mathrm{V}$: again $3\,\mathrm{A}$ and $1\,\mathrm{A}$. ✓

Example: substitution



(I) From the first example: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; inside N_1: $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

(II) A $6\,\mathrm{V}$ source fixes $v = 6\,\mathrm{V}$: again $3\,\mathrm{A}$ and $1\,\mathrm{A}$. ✓

(III) A $2\,\mathrm{A}$ source: KCL at a, $\frac{12-v}{2} = \frac{v}{6} + 2$, so $v = 6\,\mathrm{V}$: again $3\,\mathrm{A}$ and $1\,\mathrm{A}$. ✓

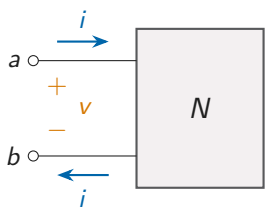
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

Ports and one-ports

Port, one-port

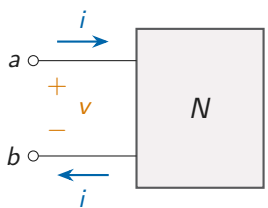
A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.



Ports and one-ports

Port, one-port

A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.

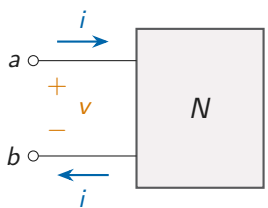


- A one-port is always a port: KCL on a closed surface around N.

Ports and one-ports

Port, one-port

A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.

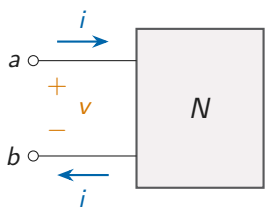


- A one-port is always a port: KCL on a closed surface around N.
- i enters the $+$ terminal of v.

Ports and one-ports

Port, one-port

A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.

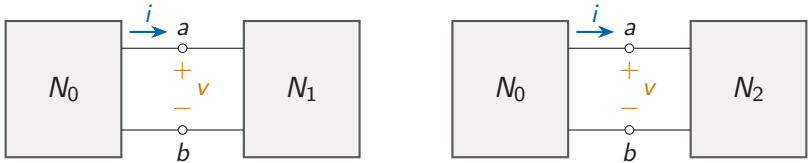


- A one-port is always a port: KCL on a closed surface around N.
- i enters the $+$ terminal of v.
- The relation between v and i is the i–v characteristic, e.g. $v = Ri$ for a resistor.

Equivalent one-ports

Equivalent one-ports

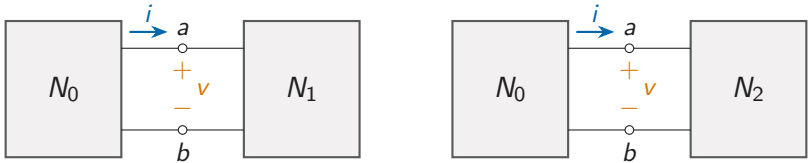
$N_1 \equiv N_2$ if they have the same i–v characteristic at their terminals.



Equivalent one-ports

Equivalent one-ports

$N_1 \equiv N_2$ if they have the same i–v characteristic at their terminals.

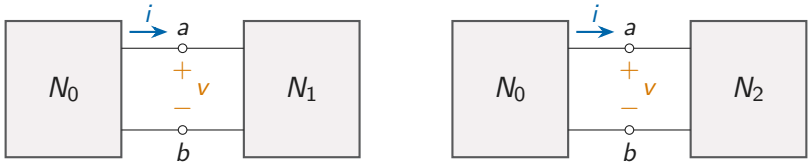


For any N_0: every branch voltage and current inside N_0 is the same in both.

Equivalent one-ports

Equivalent one-ports

$N_1 \equiv N_2$ if they have the same i–v characteristic at their terminals.



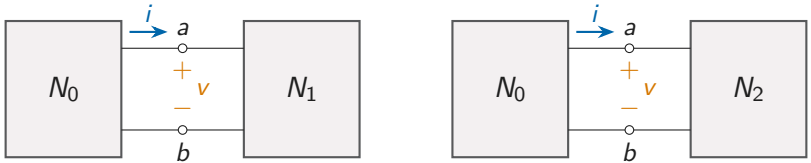
For any N_0: every branch voltage and current inside N_0 is the same in both.

Why: the equations are those of N_0 plus the i–v characteristic at the port.

Equivalent one-ports

Equivalent one-ports

$N_1 \equiv N_2$ if they have the same i–v characteristic at their terminals.



For any N_0: every branch voltage and current inside N_0 is the same in both.

Why: the equations are those of N_0 plus the i–v characteristic at the port.

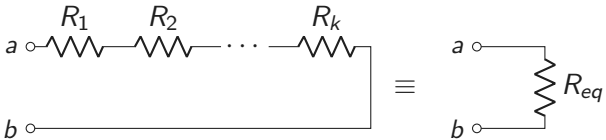
Inside N_1 and N_2 the branch values may differ.

Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same i–v characteristic at the port ✓
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

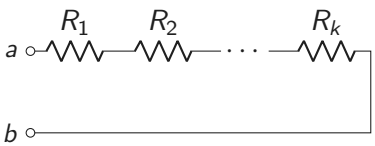
Series connection

Same current i through every element (KCL); the voltages add (KVL).

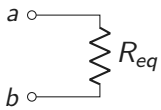


Series connection

Same current i through every element (KCL); the voltages add (KVL).



$\equiv$

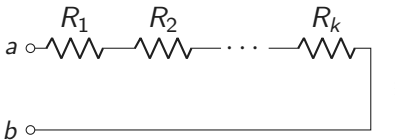


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

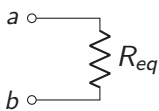
$$v = R_1 i + R_2 i + \cdots + R_k i = R_{eq} i$$

Series connection

Same current i through every element (KCL); the voltages add (KVL).



$\equiv$

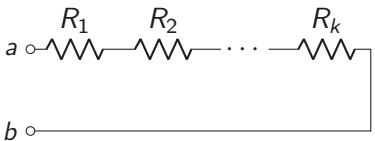


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

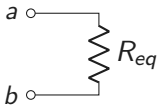
Voltage divider: the share of R_j is $v_j = \dfrac{R_j}{R_{eq}}\, v$

Series connection

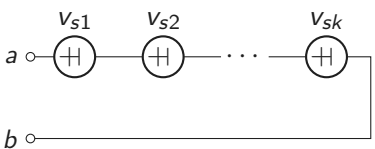
Same current i through every element (KCL); the voltages add (KVL).



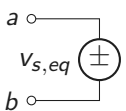
$\equiv$



$$R_{eq} = R_1 + R_2 + \cdots + R_k$$



$\equiv$

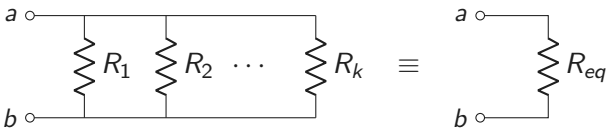


$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

All $+$ signs towards a; a source turned the other way enters with $-$.

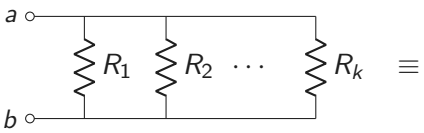
Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).



Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).

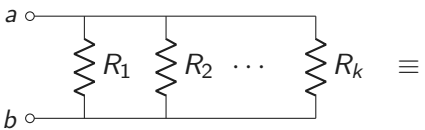


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

With $G_j = 1/R_j$: $\quad i = G_1 v + \cdots + G_k v = G_{eq} v, \quad G_{eq} = G_1 + \cdots + G_k$

Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).

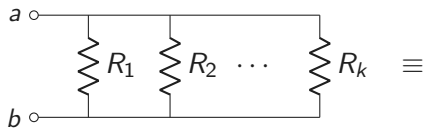


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

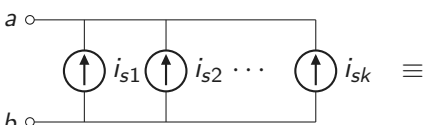
Current divider: $i_j = \dfrac{G_j}{G_{eq}}\, i.$ Two resistors: $R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).



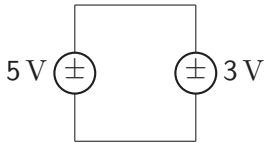
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$



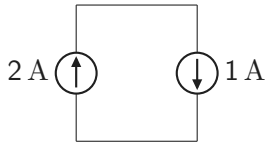
$$i_{s,eq} = i_{s1} + i_{s2} + \cdots + i_{sk}$$

All arrows towards a; a source turned the other way enters with $-$.

Two connections to avoid



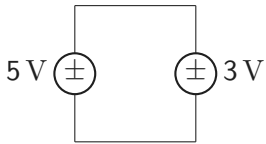
two voltage sources in parallel



two current sources in series

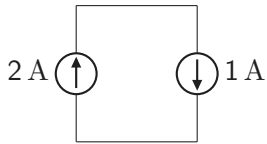
Are these allowed?

Two connections to avoid



two voltage sources in parallel

KVL fails: $5 \neq 3$.



two current sources in series

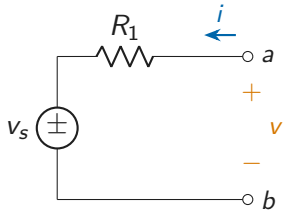
KCL fails: $2 \neq 1$.

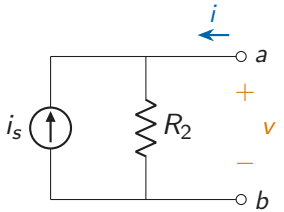
With equal values they are allowed, but the shared current (or voltage) is not fixed.

Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same i–v characteristic at the port ✓
5. **Series and parallel** — combine resistors and sources ✓
6. **One source and one resistor** — exchange a voltage source and a current source

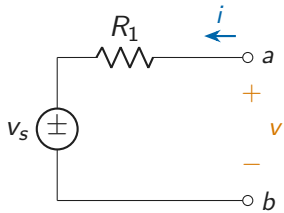
One source and one resistor

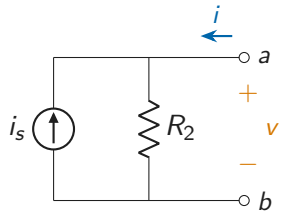




When do these two one-ports have the same i–v characteristic?

One source and one resistor

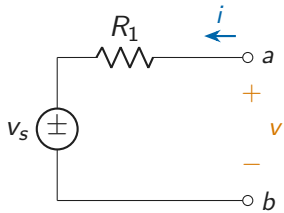




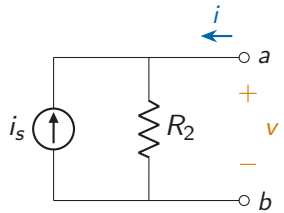
$$v = R_1 i + v_s$$

KVL around the loop: R_1 carries i.

One source and one resistor



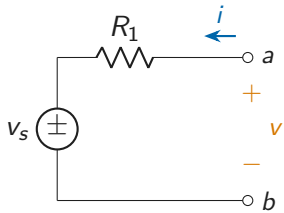
$v = R_1 i + v_s$



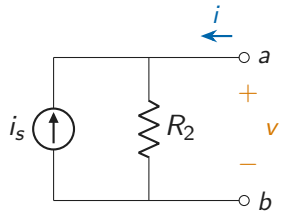
$v = R_2(i + i_s) = R_2 i + R_2 i_s$

i and i_s both flow into the top node: R_2 carries $i + i_s$.

One source and one resistor



$$v = R_1 i + v_s$$



$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

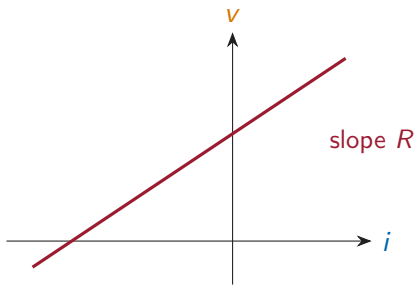
Both are straight lines in the i–v plane: equal slopes and equal intercepts.

Source transformation

Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$



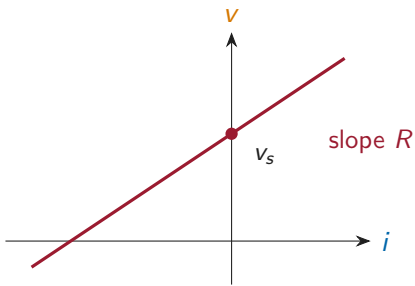
Source transformation

Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.



Source transformation

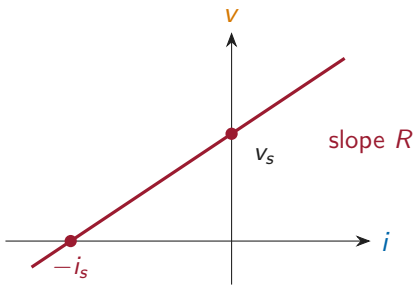
Source transformation

The two one-ports are equivalent if

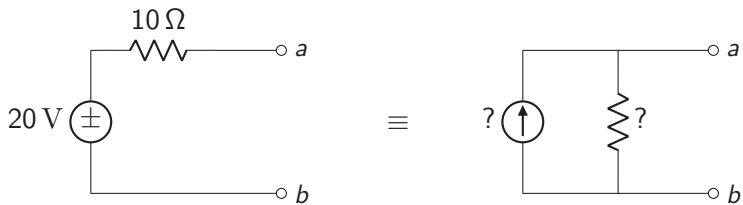
$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.

Short circuit ($v = 0$): $i = -i_s$.

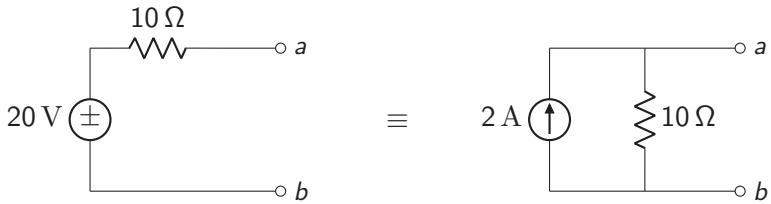


Example: a source transformation



Find the current source and the resistor.

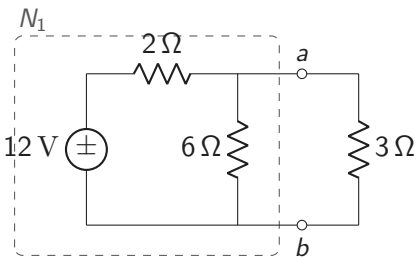
Example: a source transformation



$$R = 10\,\Omega, \qquad i_s = v_s/R = 20/10 = 2\,\mathrm{A}$$

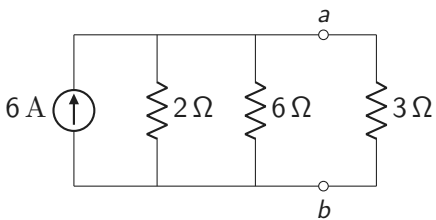
Both: open-circuit voltage $20\,\mathrm{V}$, short-circuit current $2\,\mathrm{A}$ (out of a).

Example: the first example, by transformations



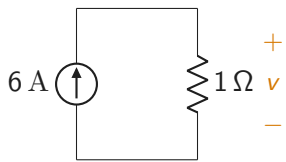
Find v again, with source transformations only.

Example: the first example, by transformations



$12\,\mathrm{V}$ with $2\,\Omega$ in series $\rightarrow$ $6\,\mathrm{A}$ with $2\,\Omega$ in parallel.

Example: the first example, by transformations

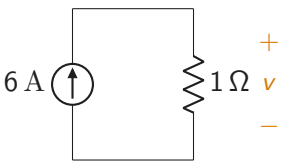


$12\,\mathrm{V}$ with $2\,\Omega$ in series $\rightarrow$ $6\,\mathrm{A}$ with $2\,\Omega$ in parallel.

$2\,\Omega$, $6\,\Omega$, $3\,\Omega$ in parallel:
$G_{eq} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3} = 1\,\mathrm{S}$, so $R_{eq} = 1\,\Omega$.

$v = 6\,\mathrm{A} \times 1\,\Omega = 6\,\mathrm{V}, \quad i = 6/3 = 2\,\mathrm{A}.$ ✓

Example: the first example, by transformations



$12\,\mathrm{V}$ with $2\,\Omega$ in series $\rightarrow$ $6\,\mathrm{A}$ with $2\,\Omega$ in parallel.

$2\,\Omega$, $6\,\Omega$, $3\,\Omega$ in parallel:
$G_{eq} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3} = 1\,\mathrm{S}$, so $R_{eq} = 1\,\Omega$.

$v = 6\,\mathrm{A} \times 1\,\Omega = 6\,\mathrm{V}, \quad i = 6/3 = 2\,\mathrm{A}.$ ✓

Careful: inside the transformed one-port the values differ. The $12\,\mathrm{V}$ source carries $3\,\mathrm{A}$, the $6\,\mathrm{A}$ source carries $6\,\mathrm{A}$.

Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.

Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.
- Two methods write the equations of the circuit as it is, without transforming it, and work for every LTI resistive circuit:
  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.

Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.
- Two methods write the equations of the circuit as it is, without transforming it, and work for every LTI resistive circuit:
  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.
- Both start from the matrices met earlier (A, M), now with linear element equations.

Summary

- LTI resistive: apart from the independent sources (external inputs), every element is linear, time-invariant and memoryless, so all equations are linear and algebraic.
- Substitution: replace N_2 by a voltage source $v_s = v$ or a current source $i_s = i$; nothing changes inside N_1.
- Equivalent one-ports: the same i–v characteristic; the rest cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- Source transformation: v_s with series $R \equiv i_s = v_s/R$ with parallel R.